Algebraic Geometry

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Lecture 1

Preliminaries

Plan

Roughly speaking, there are four parts of the course:

1. Spectrum of a commutative ring, and basics of sheaf theory.
2. Definitions of schemes and morphisms between them.
3. Properties of schemes, and properties of morphisms (replacements of naïve topological notions of compactness, Hausdorff, etc.). Basic constructions (Picard group, class group).
4. Rapid introduction to sheaf cohomology.

Prerequisites

(i) Basic undegraduate mathematics (groups, rings, topological spaces and so on).

(ii) Commutative algebra (co-requisite).

Resources

(i) Dhruv's course page.

(ii) Textbooks: Hartshorne, Vakil for Algebraic geometry, Atiyah-MacDonald for commutative algebra.

Why suffer through this? (Weil conjectures)

The Weil conjectures link two worlds. The starting data is a homogeneous polynomial $f \in \mathbb{Z}[x_0, \dots, x_n] = \mathbb{Z}[\mathbf{x}]$, of degree d.

World 1: We can consider the vanishing locus

$$X = \mathbb{V}(f) = \{(a_0 : \cdots : a_n) \mid f(a_0, \dots, a_n) = 0\} \subseteq \mathbb{P}^1_{\mathbb{C}}$$

which is a projective hypersurface. We will assume that X is smooth, i.e. there is no point $p \in X$, such that all of the partial derivatives $\frac{\partial f}{\partial x_i}$ vanish for all i. If we view $\mathbb{P}^n$ as $(\mathbb{C}^{n+1} \setminus 0)/\text{scaling}$, and $\mathbb{C}^{n+1} \setminus 0$ is given the Euclidean topology, then X is a (complex) manifold. This follows from the inverse/implicit function theorem. Moreover, X is a compact topological space. In fact, X can be triangulated.

With this, we can consider various invariants associated to X. The simplest is the Euler characteristic $\chi(X)$. For example, $\chi(\mathbb{P}^k) = k + 1$. More generally, we can consider the Betti numbers b_i, where

$$b_i = \dim_{\mathbb{Q}} H_i(X; \mathbb{Q})$$

are the dimension of the rational homology groups. In this case,

$$\chi(X) = \sum_i (-1)^i b_i$$

World 2: Fix a prime p, such that X is smooth over $\overline{\mathbb{F}_p}$ (exercise: p exists). Define N_m to be the number of points in $X\left(\mathbb{F}_{p^m}\right)$.

We can package the data of the N_ms: Consider the formal power series

$$\zeta(X,t) = \exp\left(\sum_{m=1}^{\infty} \frac{N_m}{m} t^m\right)$$

which is the *Weil ζ function*.

Theorem 0.1 (Grothendieck).

1. $\zeta(X,t)$ is a rational functon. That is,
$$\zeta(X,t) = \frac{P_0(t)\cdots P_{2n-2}(t)}{P_1(t)\cdots P_{2n-3}(t)}$$
where each $P_i(t)$ is a polynomial.
2. The degree of the polynomial $P_i(t)$ is $b_i(X)$.

Deligne proved the Riemann Hypothesis for the Weil ζ function.

1 Beyond Algebraic Varieties

1.1 Summary of affine varieties

Let $k = \overline{k}$ be an algebraically closed field. We would like to consider subsets

$$\mathbb{V}(S) \subseteq \mathbb{A}_k^n$$

where $S \subseteq k[x_1,\ldots,x_n]$. Up to isomorphism, this is the same as finitely generated k-algebras without nilpotent elements. Given $V = \mathbb{V}(S)$, we have an ideal

$$I(V) = \{f \in k[x_1,\ldots,x_n] \mid f = 0 \text{ on } V\}$$

We can then define

$$k[V] = \frac{k[x_1,\ldots,x_n]}{I(V)}$$

Conversely, given $k[V]$, a finitely generated nilpotent-free k-algebra. Choosing generators $y_1,\ldots,y_n$ of the algebra, we have a surjection $\pi : k[x_1,\ldots,x_n] \twoheadrightarrow k[V]$, sending x_i to y_i. We can then take

$$V = \mathbb{V}(\ker(\pi))$$

Lecture 2

Let $\sqrt{I}$ be the radical of an ideal I be

$$\sqrt{I} = \{f \in k[\mathbf{x}] \mid f^m \in I \text{ for some } m\}$$

Then $\mathbb{V}(S) = \mathbb{V}(\langle S\rangle) = \mathbb{V}(\sqrt{\langle S\rangle})$, where $\langle S\rangle$ is the ideal generated by S. In this case, we have a correspondence between affine varieties in $\mathbb{A}_k^n$ and radical ideals in $k[\mathbf{x}]$.

Remark 1.1. This awkwardness caused by "in affine space" can be removed, by considering varieties up to isomorphism, and coordinate rings (i.e. finitely generated nilpotent free k-algebras).

1.1.1 Basic structures

- $k[\mathbf{x}]$ is the ring of algebraic functions on $\mathbb{A}^n_k$.
- $I \subseteq k[\mathbf{x}]$ is a radical ideal.
- The coordinate ring of $V = \mathbb{V}(I)$ is $k[V] = k[\mathbf{x}]/I$.

This coordinate ring is the basic object, it's more fundamental than I.

1.1.2 Topology

Fix an affine variety $V \subseteq \mathbb{A}^n_k$. It has a Zariski topology, where the closed sets are subsets of the form

$$V \cap \mathbb{V}(S)$$

where $S \subseteq k[\mathbf{x}]$. Equivalently, we can consider the Zariski topology on $\mathbb{A}^n_k$, and then equip V with the subspace topology.

Intrinsically, we can say that $W \subseteq V$ is closed if we have that

$$W = \mathbb{V}(J)$$

where $J \subseteq k[V]$ is an ideal. Exercise: these two topologies are the same.

1.1.3 Nullstellensatz

That is, what are the maximal ideals in $k[\mathbf{x}]$? Equivalently, in a nilpotent free finitely generated k-algebra $k[V]$?

Fix $V = \mathbb{V}(I)$, and pick $p \in V$. There is a natural ring homomorphism $\mathrm{ev}_p : k[V] \to k$, given by $\mathrm{ev}_p(f) = f(p)$.

> **Claim 1.2.** ev_p is surjective.

Proof. Exercise. □

The kernel of ev_p is a maximal ideal in $k[V]$. Therefore, points on V are one source of maximal ideals.

> **Theorem 1.3** (Nullstellensatz)**.** There is a bijection
>
> $$\{\text{points in } V\} \leftrightarrow \{\text{maximal ideals in } k[V]\}$$

1.1.4 Functions

Each $f \in k[V]$ gives a function $V \to k (= \mathbb{A}^1_k)$, sending p to $f(p)$. Equivalently, if we denote $\mathfrak{m}_p = \ker(\mathrm{ev}_p)$, then this is the quotient map

$$k[V] \twoheadrightarrow k = k[V]/\mathfrak{m}_p$$

1.1.5 Morphisms

Given $V \subseteq \mathbb{A}^n_k$, $W \subseteq \mathbb{A}^m_k$, morphisms $V \to W$ are constructed as follows.

1. Take functions $f_1, \ldots, f_m \in k[x_1, \ldots, x_n]$.
2. This gives a map $\varphi = (f_1, \ldots, f_m) : V \to \mathbb{A}^m_k$.
3. A morphism $V \to W$ must additionally satisfy that $\varphi(V) \subseteq W$.

There is a more intrinsic construction. Any such $\varphi : V \to W$ as above, gives rise to a pullback map $\varphi^* : k[W] \to k[V]$, where $\varphi^*(g) = g \circ \varphi$.

There is a bijection

$$\{\text{Morphisms } V \to W\} \leftrightarrow \{k\text{-algebra homomorphisms } k[W] \to k[V]\}$$

In practice, the right hand side is much cleaner than the right hand side.

Lesson: We ♡ coordinate rings.

Upshot: Let R be a finitely generated, nilpotent free k-algebra, the topological space

$$X_R = \{\mathfrak{m} \subseteq R \mid \mathfrak{m} \text{ is a maximal ideal}\}$$

Closed sets are given by $\mathbb{V}(S)$, for $S \subseteq R$.

1.2 Limitations

Question 1.4. What is an abstract (not necessarily affine) variety?

Morally, it's an object which locally looks like an affine variety.

Example 1.5 (Non-algebraically closed fields)

Consider the ideal

$$I = \left\langle x^2 + y^2 + 1 \right\rangle \in \mathbb{R}[x, y]$$

Then $\mathbb{V}(I) = \varnothing \subseteq \mathbb{R}^2$. I is a prime ideal, therefore radical, so the naïve Nullstellensatz fails.

Question 1.6. Given a commutative ring R, is there a natural topological space X_R, on which R is the ring of functions? For example, $R = \mathbb{Z}, \mathbb{Z}[x]$ and so on.

Example 1.7 (Why restrict to radical ideals)

Assume $k = \overline{k}$.

$$C = \mathbb{V}(y - x^2) \subseteq \mathbb{A}^n_k$$

and

$$D = \mathbb{V}(y)$$

Then

$$C \cap D = \mathbb{V}(y, y - x^2) = \mathbb{V}(x^2, y) = \{(0, 0)\}$$

But $\left\langle y, y - x^2 \right\rangle$ has radical $\langle x, y \rangle$, it is not equal to the ideal $\langle x, y \rangle$. The intersection wants to be two points. Take

$$D_\delta = \mathbb{V}(y + \delta)$$

where $\delta \in k \setminus 0$, then $C \cap D_\delta$ has two points.

1.3 Spectrum of a ring

Basic idea: Given a ring A, we will define

$$\operatorname{Spec}(A) = \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ a prime ideal}\}$$

Why prime?

- Functoriality - $A \to B$ induces $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ by pullback.
- Some rings have very few maximal ideals.

Let A be a commutative ring,
Goal: Define a topological space X_A on which A is naturally the ring of functions.

Definition 1.8 (Zariski spectrum)
The Zariski spectrum of A is

$$\mathrm{Spec}(A) = \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ a prime ideal}\}$$

Remark 1.9. We might also consider the maximal ideal spectrum, but this fails to be functorial. That is, a ring homomorphism $\varphi : A \to B$ induces a map $\varphi^* = \varphi^{-1} : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$, by considering the preimage. On the other hand, no such map exists for maximal ideals.

Why? Whatever X_A could be, given a point $p \in X_A$, we should have an evaluation map

$$\begin{aligned} \mathrm{ev}_p : A &\to K_p \\ f &\mapsto f(p) \end{aligned}$$

where K_p is a field. Note we allow the field K_p to depend on p. Nor are we requiring that ev_p be surjective. The image of ev_p is an integral domain, and moreover, any map to an integral domain R is also a map to a field $F = \mathrm{Frac}(R)$.

Subject to this, homomorphisms $f : A \to K$, with K a field, corresponds to $\ker(f)$, which is a prime ideal.

Example 1.10
Taking $A = \mathbb{Z}$, we have that

$$\mathrm{Spec}(\mathbb{Z}) = \{\langle p \rangle \mid p \in \mathbb{N} \text{ a prime}\} \cup \{0\}$$

Pick a function $123 \in \mathbb{Z}$. Now evaluate at different points, which is just reduction modulo p, i.e.

$$\begin{aligned} 123(2) &= 1 \in \mathbb{Z}/2 \\ 123(3) &= 0 \in \mathbb{Z}/3 \\ 123(0) &= 123 \in \mathbb{Z} \subseteq \mathbb{Q} \end{aligned}$$

Example 1.11
Taking $A = \mathbb{R}[x]$, $\mathrm{Spec}(A)$ has elements corresponding to

1. irreducible quadratics in x, e.g. $\langle x^2 + 1 \rangle$, which corresponds to $(\mathbb{C}\backslash\mathbb{R})$/complex conjugation.
2. linear polynomials in x, e.g. $\langle x - 2 \rangle$, corresponding to $\mathbb{R}$,
3. 0.

Hence $\mathrm{Spec}(A)$ is the upper half plane, plus the point zero.

Definition 1.12
If A is an integral domain, then $0 \in \mathrm{Spec}(A)$ is called the generic point.

Remark 1.13. The value of $f \in A$ at $\mathfrak{p} \in \operatorname{Spec}(A)$ is $f + \mathfrak{p} \in A/\mathfrak{p}$.

1.4 Topology on $\operatorname{Spec}(A)$

Motivation: In the Zariski topology, the zero sets should be the vanishing sets of functions. Fix $f \in A$, define the vanishing set of f to be

$$\mathbb{V}(f) = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid f \in \mathfrak{p}\} = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid f(\mathfrak{p}) = 0 \in A/\mathfrak{p}\}$$

More generally, if $I \subseteq A$ is an ideal, then

$$\mathbb{V}(I) = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid I \subseteq \mathfrak{p}\} = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid f(\mathfrak{p}) = 0 \text{ for all } f \in I\}$$

Definition 1.14

The sets $\mathbb{V}(I) \subseteq \operatorname{Spec}(A)$, for $I \subseteq A$ ideals, satisfy the requirements for closed sets in a topological space.

Proof. Easy to see $\varnothing$ and $\operatorname{Spec}(A)$ are closed. Moreover,

$$\bigcap_\alpha \mathbb{V}(I_\alpha) = \mathbb{V}\left(\sum_\alpha I_\alpha\right)$$

and

$$\mathbb{V}(I_1) \cup \mathbb{V}(I_2) = \mathbb{V}(I_1 \cap I_2)$$

For this, $\subseteq$ is clear. For the reverse containment,

$$I_1 I_2 \subseteq I_1 \cap I_2$$

If $\mathfrak{p} \supseteq I_1 \cap I_2$, then it contains $I_1 I_2$. Since $\mathfrak{p}$ is prime, then either I_1 or I_2 is contained in $\mathfrak{p}$. □

Example 1.15

Let $k = \overline{k}$, and $A = k[x, y]$. Then

$$\operatorname{Spec}(A) \supseteq \operatorname{maxSpec}(A) = k^2$$

where $\operatorname{maxSpec}(A)$ is the set of maximal ideals, which by the Nullstellensatz corresponds to the points in $\mathbb{A}^2_k$. There is a point for each non-maximal prime ideal, e.g.

$$\langle x - y \rangle \quad \langle y^2 - x^3 \rangle \quad \langle y - x^2 \rangle$$

Consider the generic point $0 \in \operatorname{Spec}(A)$. Then the closure of $\{0\}$ is $\operatorname{Spec}(A)$. On the other hand, the closure of a maximal ideal is itself.

Now consider the ideal $\langle f \rangle$, where f is an irreducible polynomial, e.g. $y^2 - x^3$. The closure of $\{\langle f \rangle\}$ contains every maximal ideal $\langle x - a, y - b \rangle$ such that $b^2 - a^3 = 0$, since such maximal ideals contain $\langle f \rangle$.

Consider the ring $A = \mathbb{C}$, the ring of formal power series in x. Then

$$\operatorname{Spec}(A) = \{\langle 0 \rangle, \langle x \rangle\}$$

which

$$\overline{\{\langle 0 \rangle\}} = \operatorname{Spec}(A)$$
$$\overline{\{\langle x \rangle\}} = \{\langle x \rangle\}$$

On the other hand, maxSpec(A) is a single (closed) point, which is the same as Spec($\mathbb{C}$).

Spec(A) is a local model of a scheme. Last time, we considered Spec(A) as a set, and as a topological space.

1.5 Functions on open sets in Spec(A)

Definition 1.16
Let $f \in A$. The distinguished open corresponding to f is

$$U_f = F(f) = \mathrm{Spec}(A) \setminus \mathbb{V}(f)$$

Lemma 1.17. The distinguised open sets form a base for the Zariski topology on Spec(A).

Proof. Examples sheet 1. □

Starting point: We are already thinking about A as "functions on Spec(A)".

Lemma 1.18. The subspace $U_f \subseteq \mathrm{Spec}(A)$ is (naturally) homeomorphic to Spec(A_f), where A_f is the localisation of A at f.

Aside on localisation: $A_f = A[t]/\langle tf - 1\rangle$. For example, if $A = \mathbb{C}[x]$ and $f = x$, then $A_f = \mathbb{C}[x, x^{-1}]$. There is an obvious map $A \to A_f$, however if f is a zero divisor, this map may not be injective.

Since we have a ring homomorphism $A \to A_f$, we have an induced map $\mathrm{Spec}(A_f) \to \mathrm{Spec}(A)$.

Proof. Primes in A_f are exactly in bijection with primes in A that do not contain f. Why? Let $\theta : A \to A_f$ be the natural map.

Claim 1.19. Every ideal in A_f is generated by $\theta(I)$ for some ideal $I \subseteq A$.

For details, see Atiyah-MacDonald, or Commutative Algebra. □

Example 1.20
Let $A = \mathbb{C}[x, y]$, $f = xy$. Then

$$U_f = \mathbb{C}^2 \setminus (\{x\text{-axis}\} \cup \{y\text{-axis}\}) \cup \{\text{weird, i.e. non-closed, points}\}$$

A more algebraic way of thinking about this is that

$$A_f = \frac{\mathbb{C}[x, y][t]}{\langle txy - 1\rangle} \cong \mathbb{C}[x, x^{-1}, y, y^{-1}]$$

If a prime ideal $\mathfrak{p}$ in $\mathbb{C}[x, y]$ contains x or y, then the ideal generated by the image of $\mathfrak{p}$ is the unit ideal.

This sets up the following association:

$$\begin{aligned} \text{Distinguished open set} &\to \text{Rings} \\ U_f &\mapsto A_f \end{aligned}$$

Observe: If $U_{f_1} \subseteq U_{f_2}$, there is a natural ring homomorphism $A_{f_2} \to A_{f_1}$.

2 Sheaves

Sheaves will formalise behaviour of the assignment above.

2.1 Presheaves

Definition 2.1 (presheaf)

A *presheaf $\mathcal{F}$ of abelian groups* on a topological space X is an assignment

$$\begin{aligned} \text{Open sets in } X &\to \text{Abelian groups} \\ U &\mapsto \mathcal{F}(U) \end{aligned}$$

such that if $U \subseteq V$, then there is a homomorphism $\operatorname{res}_U^V : \mathcal{F}(V) \to \mathcal{F}(U)$, with

- $\operatorname{res}_U^U = \mathrm{id}_{\mathcal{F}(U)}$,
- if we have $U \subseteq V \subseteq W$ opens, then

$$\operatorname{res}_U^V \circ \operatorname{res}_V^W = \operatorname{res}_U^W$$

Keep in mind: If X is any topological space, then

$$U \mapsto \{f : U \to \mathbb{R} \mid f \text{ is continuous}\}$$

is a presheaf.

Remark 2.2. $\mathcal{F}$ is just a functor between certain categories. It is easy to define presheaves of sets, rings, etc.

Morphisms of presheaves

Let $\mathcal{F}, \mathcal{G}$ be presheaves on a topologial space X.

Definition 2.3

A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is, for each open set U, a homomorphism

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

compatible with restriction. That is, if $V \subseteq U$ are open, then

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \operatorname{res}_V^U} & & \downarrow{\scriptstyle \operatorname{res}_V^U} \\ \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \end{array}$$

commutes.

2.2 Sheaves - Definitions and Examples

Definition 2.4 (Sheaf)

A sheaf $\mathcal{F}$ on X is a presheaf such that

S1. If $U \subseteq X$ is open, and $\{U_i\}$ is an open cover of U, and if $s \in \mathcal{F}(U)$ is such that

$$s|_{U_i} = \operatorname{res}_{U_i}^U(s) = 0$$

for all i, then $s = 0$.

S2. If U and $\{U_i\}$ are as above, given $s_i \in \mathcal{F}(U_i)$, with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, then there exists $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all i.

Definition 2.5

A morphism of sheaves is a morphism of the underlying presheaves.

Example 2.6 (Non-example)

Let $X = \mathbb{C}$ with the Euclidean topology. Set

$$\mathcal{F}(U) = \{f : U \to \mathbb{C} \mid f \text{ bounded analytic}\}$$

This is *not* a sheaf, since it does not satisfy S2.

Example 2.7 (Non-example)

Fix an abelian group G, and set $\mathcal{F}(U) = G$ for all U. This is a presheaf, but not a sheaf. If U_1, U_2 are disjoint, then the sheaf axioms force $\mathcal{F}(U_1 \cup U_2) = G \times G$.

Lecture 5

Example 2.8 (Constant sheaf)

Fix an abelian group G, define the constant sheaf with values in G to be

$$\mathcal{F}(U) = \{f : U \to G \mid f \text{ locally constant on } U\}$$

In practice, if U is connected, then $\mathcal{F}(U) = G$. Otherwise, $\mathcal{F}(U)$ is $G^{\#\text{connected components}}$. In particular, if we take $G = \mathbb{Z}$, then this sheaf is denoted $\underline{\mathbb{Z}}$, and

$$\underline{Z}(X) = \mathbb{Z}^{\#\text{connected components}} = \mathrm{H}_0(X, \mathbb{Z})$$

This leads to the question, what about the other (co)homology groups?

Example 2.9 (Structure sheaf)

Let V be an irreducible affine variety on $k = \overline{k}$. Define a sheaf $\mathcal{O}_V$ as follows,

$$\mathcal{O}_V(U) = \{f : k(V) \mid f \text{ is regular at all } p \in U\}$$

This is called the structure sheaf. Note $k(V) = \mathrm{Frac}(k[V])$.

2.3 Basic constructions

Fix a sheaf $\mathcal{F}$ on a topological space X.

Terminology: A section of $\mathcal{F}$ over $U \subseteq X$ is an element of $\mathcal{F}(U)$. The group of global sections is $\mathcal{F}(X)$.

Definition 2.10 (Stalk)

Fix a point $p \in X$. Then the stalk of $\mathcal{F}$ at p, denoted $\mathcal{F}_p$, is

$$\mathcal{F}_p = \frac{\{(s, U) \mid s \in \mathcal{F}(U), p \in U\}}{\sim}$$

where $(s, u) \sim (s', u')$ if there exists a nonempty open $p \in W \subseteq U \cap U'$, such that $s|_W = s'|_W$.

$\mathcal{F}_p$ is sections of $\mathcal{F}$ on some neighbourhood of p.

Group structure on $\mathcal{F}_p$

Given $(s, U), (t, V)$ representing elements of $\mathcal{F}_p$, we can define

$$(s, U) + (t, V) = (s|_W + t|_W, W)$$

for any neighbourhood $W \subseteq U \cap V$ of p.

Example 2.11

Consider the structure sheaf $\mathcal{O}_{\mathbb{A}^1_k}$, where $k = \overline{k}$. Open sets of $\mathbb{A}^1_k$ are complements of finite sets (and the empty set).

- $\mathcal{O}_{\mathbb{A}^1_k} = k[x]$.
- $U = \mathbb{A}^1_k \setminus \{1\}$, then $\mathcal{O}_{\mathbb{A}^1_k}(U) = k[x, (x-1)^{-1}]$ (c.f. localisation).

Therefore, $\mathcal{O}_{\mathbb{A}^1_k, 0} = \{f(x)/g(x) \mid g(x) \neq 0\}$, is the local ring of $\mathbb{A}^1_k$ at 0.

Proposition 2.12. Let $\mathcal{F}, \mathcal{G}$ be sheaves on X. Let $f : \mathcal{F} \to \mathcal{G}$ be a morphism. If the induced maps $f_p : \mathcal{F}_p \to \mathcal{G}_p$ is an isomorphism for all $p \in X$, then $\mathcal{F}$ is an isomorphism.

Exercise: The induced map makes sense.

Proof. We will show that for every open $U \subseteq X$, the morphism $f_U : \mathcal{F}(U) \to \mathcal{G}(U)$ is an isomorphism. Given this, we will define the inverse map $f^{-1} : \mathcal{G} \to \mathcal{F}$ to be f_U^{-1} on each U.

Exercise: This is compatible with gluing.

Injectivity. Suppose $s \in \mathcal{F}(U)$, with $f_U(s) = 0$. This determines elements $s_p \in \mathcal{F}_p$ for all $p \in U$. These are all zero, since $f_p(s_p) = 0 \in \mathcal{G}_p$, and f_p is injective. There exists open sets U_p around each $p \in U$, such that $s|_{U_p} = 0$, by S2. But then by S1., $s = 0$.

Surjectivity. Let $t \in \mathcal{G}(U)$. We will build $s \in \mathcal{F}(U)$ with $f_U(s) = t$. Let $t_p \in \mathcal{G}_p$ for the image of t. Denote $s_p = f_p^{-1}(t_p)$. This makes sense since f_p is an isomorphism. Note that *a priori*, we don't know that the s_p give us an element $s \in \mathcal{F}(U)$.

Find a representative $(s_{V(p)}, V(p))$ of s_p, with $V(p)$ a neighbourhood of p, such that

$$f_{V(p)}((s_{V(p)}, V(p))) \sim (t, U)$$

Why do the $s_{V(p)}$ glue over different choices of p? They glue because we have already shown injectivity of f_U. □

Remark 2.13. (i) Even easier than the proposition, there is a natural map

$$\mathcal{F}(U) \to \prod_{p \in U} \mathcal{F}_p$$

and the map is injective.

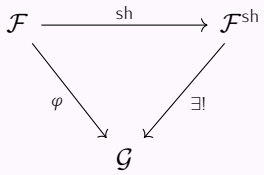
(ii) Given $\varphi, \psi : \mathcal{F} \to \mathcal{G}$, with $\varphi_p = \psi_p$ for all $p \in X$. Then $\varphi = \psi$.

Lecture 6

Sheafification

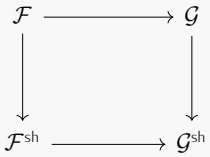
"The best way of turning a presheaf into a sheaf"

Definition 2.14 (sheafification)
Let $\mathcal{F}$ be a presheaf on X. A morphism $\mathrm{sh} : \mathcal{F} \to \mathcal{F}^{\mathrm{sh}}$ is a sheafification if $\mathcal{F}^{\mathrm{sh}}$ is a sheaf, and for any morphism $\varphi : \mathcal{F} \to \mathcal{G}$, with $\mathcal{G}$ a sheaf, there exists a unique commuting diagram



Remark 2.15. (i) If $\mathrm{sh} : \mathcal{F} \to \mathcal{F}^{\mathrm{sh}}$ exists, then it is unique up to unique isomorphism.

(ii) Functoriality. Presheaf morphisms induce morphisms of sheaves. That is, we have that



commutes.

Note that if $\mathcal{F}$ is a presheaf on X, then it has stalks $\mathcal{F}_p$ for each $p \in X$, and an element of $\mathcal{F}_p$ is called a germ of $\mathcal{F}$ at p.

Proposition 2.16. Let $\mathcal{F}$ be a presheaf on X. Define

$$\mathcal{F}^{\mathrm{sh}}(U) = \big\{(f_p)_{p\in U} \mid f_p \in \mathcal{F}_p \text{ and for every } p, \text{ there exists an open } V_p \subseteq U \text{ neighbourhood of } p, \\ \text{and a section } s \in \mathcal{F}(V_p) \text{ such that } s_q = f_q \in \mathcal{F}_q \text{ for all } q \in V_p\big\}$$

We are doing two things here.

1. We are adding sections, coming from coherent germs.
2. Throwing away sections that are incoherent.

Proof. The restriction maps are clear. The map $\mathcal{F} \to \mathcal{F}^{\mathrm{sh}}$ is the obvious one, namely $s \mapsto (s_p)_{p\in U}$.

Exercise: Verify the universal property. A hint for the exercise: if we have a sheaf $\mathcal{G}$ and a morphism $f : \mathcal{F} \to \mathcal{G}$. On an open set U, if we have a section $s \in \mathcal{F}(U)$, that restricts to zero on an open cover, then its image in $\mathcal{G}(U)$ is zero, as $\mathcal{G}$ is a sheaf. □

Corollary 2.17 (of the construction). If $\mathcal{F}$ is a presheaf on X, then $\mathcal{F}$ and $\mathcal{F}^{\mathrm{sh}}$ have the same stalks at every $p \in X$.

Exercise (stupid): Find a nonzero presheaf $\mathcal{F}$ on X, so that $\mathcal{F}^{\mathrm{sh}} = 0$.

2.4 Kernels, cokernels, etc.

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves on X. A presheaf kernel/image/cokernel assigns U to

$$\ker(\varphi_U), \mathrm{im}(\varphi_U), \mathrm{coker}(\varphi_U)$$

respectively.
Exercise: Suppose $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves. Then the presheaf kernel of φ is a sheaf.
On the other hand, the image and cokernels need not be sheaves.

Example 2.18

Let $X = \mathbb{C}$ with the usual Euclidean topology. Consider the sheaf $\mathcal{O}_X$ of holomorphic functions under addition. Another sheaf is $\mathcal{O}_X^*$, whose abelian groups are

$$\mathcal{O}_X^*(U) = \{f \text{ holomorphic on } U \text{ and nonwhere zero}\}$$

under multiplication.
Key point: Exponentiation makes things nonzero, and logarithms exists only locally.
Define the exponentiation map

$$\begin{aligned} \exp : \mathcal{O}_X(U) &\to \mathcal{O}_X^*(U) \\ \exp(f) &= e^f \end{aligned}$$

Do all sections of $\mathcal{O}_X^*$ arise in this way? Namely by exponentiation. Take

$$\begin{aligned} U_1 &= \mathbb{C} \setminus [0, \infty) \\ U_2 &= \mathbb{C} \setminus (-\infty, 0] \end{aligned}$$

Take $U = U_1 \cup U_2 = \mathbb{C} \setminus \{0\}$. Take a function $z \in \mathcal{O}_X^*(U)$. This lies in the presheaf cokernel of exp, because the logarithm does not exists on U. However, on U_1, U_2, log exists. So the cokernel is trivial.

In conclusion, the image of exp as a presheaf morphism is not a sheaf.

We'll see next time since the sheafification of

$$\operatorname{coker}(\exp : \mathcal{O}_X \to \mathcal{O}_X^*)$$

is 0, the exponential "should be" a surjective sheaf morphism.

Definition 2.19

For a morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of sheaves, the sheaf *cokernel (image)* is the sheafification of the presheaf cokernel (image).

A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is injective if $\ker(\varphi) = 0$, and it is surjective if $\operatorname{im}(\varphi) = \mathcal{G}$.

Remark 2.20. The sequence

$$0 \to \underline{2\pi i\mathbb{Z}} \to \mathcal{O}_X \to \mathcal{O}_X^* \to 0$$

is exact as sheaves. In fact, this is true for any complex manifold X.